

$^{37}\text{Cl}(\text{p,t})$ 1971Vi02,2017Ch22

$^{37}\text{Cl} \rightarrow ^{35}\text{Cl} + 2\text{n}$ from $J^\pi = 3/2^+$ ^{37}Cl ground state.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. Targets were natural purified chlorine gas 100 mm in diameter and 25 mm in height and solid AgCl of isotopic ^{37}Cl . The reaction particles were detected using two separate counter telescopes consisting of a 200- μm phosphorous-drifted silicon ΔE detector, a 3-mm lithium-drifted silicon E detector, and a 3-mm lithium-drifted silicon E-reject detector with FWHM=130 keV for tritons. Measured $\sigma(E_t, \theta)$. Deduced L-transfers for 7 levels from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

2017Ch22: a 30-MeV proton beam was produced from the Holifield Radioactive Ion Beam Facility at Oak Ridge National Laboratory. The target was $\sim 600 \mu\text{g}$ PbCl_2 ($\sim 95\%$ enriched and $\sim 160 \mu\text{g}/\text{cm}^2$ in ^{37}Cl) on a $20 \mu\text{g}/\text{cm}^2$ parylene (C_8H_8) backing. Tritons were detected using the annular silicon detector array (SIDAR) consisting of 100- μm - ΔE and 1000 μm -E telescopes with FWHM=60-100 keV. Measured $\sigma(E_t, \theta)$. Deduced levels, J, π , and L-transfers from TWOFNR-DWBA analysis of the measured $\sigma(\theta)$. Provided the first spin-parity constraint of a 306-keV $^{34}\text{S}(\text{p}, \gamma)$ resonance potentially important for nova nucleosynthesis.

Other: **1985Mi06**.

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>L[‡]</u>	Comments
0	0	L: from 2017Ch22 . Other: 0+2 in 1971Vi02 . Relative peak intensity=100 (1971Vi02).
1220 40		Relative peak intensity=0 (1971Vi02).
1750 40	2	Relative peak intensity=41 (1971Vi02).
2650 40	2	Relative peak intensity=41 (1971Vi02). Unresolved 2645 and 2694 doublet.
3000 40		Relative peak intensity=5 (1971Vi02).
5650 50	0+2	Relative peak intensity=22 (1971Vi02).
6677 15	(2)	E(level): from 2017Ch22 . 2017Ch22 adopts L=2 and states that combinations of L=0+2 or L=1+3 are also possible. Because L=2 and L=0 are better fits than L=1 and L=3, a positive parity assignment for the 6677 level is strong.
7250 50		Relative peak intensity=9 (1971Vi02).

[†] From **1971Vi02**, unless otherwise noted.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in **1971Vi02**, unless otherwise noted.